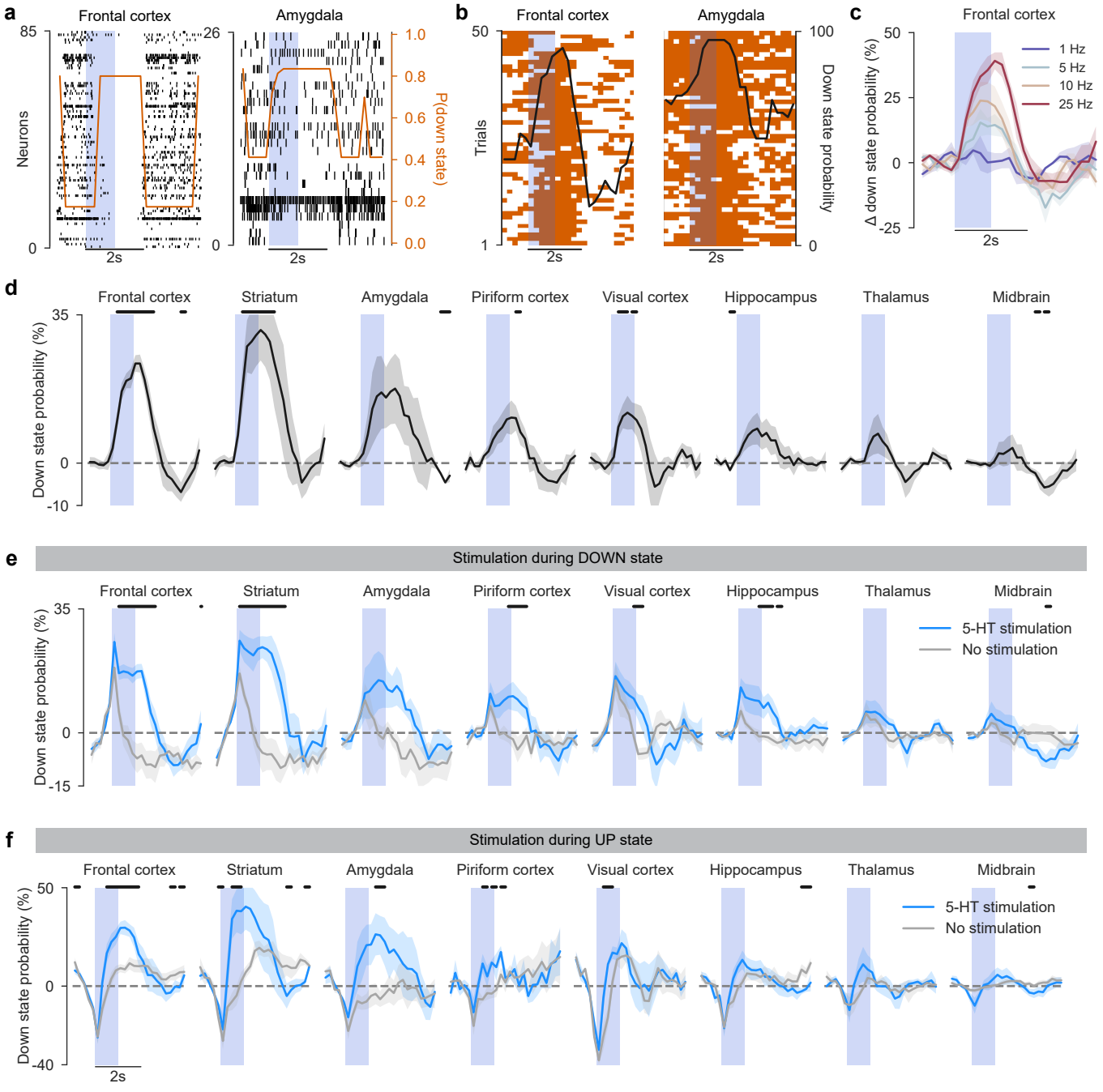

a
Frontal cortex
Amygdala
Neurons
85
0
26
0
2s
2s
P(down state)
1.0
0.8
0.6
0.4
0.2
0.0
b
Frontal cortex
Amygdala
Trials
50
1
100
0
Down state probability
2s
2s
c
Frontal cortex
Δ down state probability (%)
50
25
0
-25
1 Hz
5 Hz
10 Hz
25 Hz
2s
d
Frontal cortex
Striatum
Amygdala
Piriform cortex
Visual cortex
Hippocampus
Thalamus
Midbrain
Down state probability (%)
35
0
-10
e
Stimulation during DOWN state
Frontal cortex
Striatum
Amygdala
Piriform cortex
Visual cortex
Hippocampus
Thalamus
Midbrain
Down state probability (%)
35
0
-15
5-HT stimulation
No stimulation
f
Stimulation during UP state
Frontal cortex
Striatum
Amygdala
Piriform cortex
Visual cortex
Hippocampus
Thalamus
Midbrain
Down state probability (%)
50
0
-40
2s
5-HT stimulation
No stimulation